

2(a)	(gravitational) field strength equals (gravitational) potential gradient	M1
	reference to minus sign	A1
2(b)(i)	potential is zero at infinity	B1
	(gravitational) force is attractive	B1
	(test) mass getting closer (from infinity) loses potential energy	B1
2(b)(ii)	• potential at (surface of) planet is smaller than at (surface of) moon • potential gradient at (surface of) planet is smaller than at (surface of) moon • magnitude of potential varies inversely with distance from centre near the spheres • (point of) maximum potential is nearer to moon than planet <i>Any two points, 1 mark each</i>	B2
2(b)(iii)	sketch: one curve, starting with gradient of decreasing magnitude at $2R$ and finishing with gradient of increasing magnitude at $D - R$	B1
	field strength shown as zero (only) near the point of maximum potential	B1
	negative field strength near one sphere and positive field strength near the other	B1